Jarzynski equality for a pulled polymer in 2D

Theory companion: model, fluctuation theorems, estimators and numerical scheme

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Abstract

This document gives the theory behind the simulations of the companion report (`report.pdf`). It contains the model of an anchored 2D bead–spring chain pulled by a moving harmonic trap in a Langevin bath. It derives the Jarzynski equality in three settings (Hamiltonian, discrete-time Markov, continuous-time Langevin) and explains what the equality says about the free energy. It derives the Crooks fluctuation theorem, and every estimator of ΔF used in the report: one-sided Jarzynski with its bias and variance, cumulant expansions, effective sample size, reverse Jarzynski, and Bennett's acceptance ratio. It also describes the exact free-energy references and the numerical scheme. The first version of this document was written before any code. This version incorporates the design decisions taken and two corrections found while running the simulations (Sec. 11).

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1 Comments on the original brief

Three points where the brief needed adjusting

1. **Random forces alone are not enough.** A bath has to do two things: kick the beads (noise) *and* slow them down (friction). With kicks only, every kick adds energy on average, so $\langle v^2 \rangle$ grows linearly in time and the chain heats up without bound. Noise and friction come from the same molecular collisions, and their strengths are tied by the fluctuation–dissipation relation $\sigma^2 = 2\gamma m k_\mathrm{B} T$ (Sec. 3).

2. **What fluctuates is the work, not the free energy.** $\Delta F = F(\boldsymbol{\lambda}_1) - F(\boldsymbol{\lambda}_0)$ is a property of two *equilibrium* states, so it is one number. The *work* W done by the puller differs on every repetition; its distribution $P(W)$ is what changes with T, N and speed. Jarzynski extracts the single number ΔF from the distribution of irreversible works.

3. **How the chain is pulled.** Instead of moving the last bead along a prescribed path (a constraint), the last bead is held in a harmonic trap whose centre $\boldsymbol{\lambda}(t)$ moves, as in optical-tweezer and AFM experiments. The work is then an explicit formula (Eq. 12).

2 Physical model

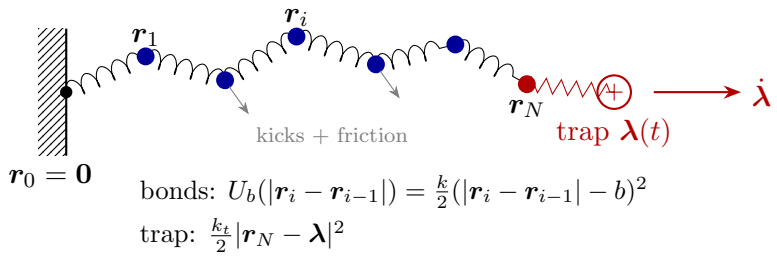


Figure 1: The system. Bead 0 is fixed at the origin; beads $1 \dots N$ move in the plane and exchange energy with the bath; bead N is held by a harmonic trap whose centre moves from $\boldsymbol{\lambda}_0$ to $\boldsymbol{\lambda}_1$ in a time τ.

The state is $\boldsymbol{x} = (\boldsymbol{r}_1, \dots, \boldsymbol{r}_N; \boldsymbol{p}_1, \dots, \boldsymbol{p}_N) \in \mathbb{R}^{4N}$, with $\boldsymbol{r}_0 \equiv \mathbf{0}$. The Hamiltonian depends on the control parameter $\boldsymbol{\lambda} \in \mathbb{R}^2$:

$$H(\boldsymbol{x}; \boldsymbol{\lambda}) = \sum_{i=1}^{N} \frac{|\boldsymbol{p}_i|^2}{2m} + \sum_{i=1}^{N} U_b\big(|\boldsymbol{r}_i - \boldsymbol{r}_{i-1}|\big) + \frac{k_t}{2}\,|\boldsymbol{r}_N - \boldsymbol{\lambda}|^2, \qquad U_b(r) = \frac{k}{2}(r-b)^2. \tag{1}$$

N is the number of mobile beads, which equals the number of bonds.

One model, two limits.

- $b = 0$, **Gaussian (Rouse) chain**: all forces are linear, so the free energy, the dynamics and the full work distribution are known exactly (Sec. 9.1). The elasticity is *energetic*. Role: validation.
- $b = 1$, stiff bonds ($kb^2 \gg k_\mathrm{B}T$), **nearly freely-jointed chain**: bond lengths barely change, so pulling works against *entropy* (rubber elasticity, restoring force $\propto T$). Role: the physics model.

Simplifications. The chain is a phantom chain (no excluded volume, which is what makes the exact reference of Sec. 9.2 possible), it has no bending stiffness, and the bath is free-draining (no hydrodynamic interactions).

Protocol. $\boldsymbol{\lambda}(t) = \big(x_0 + (x_1 - x_0)t/\tau,\, 0\big)$ with $x_0 = 0.2L$, $x_1 = 0.7L$, where $L = Nb$ is the contour length ($L = 0.3N$ for $b = 0$). The chain starts in equilibrium at $\boldsymbol{\lambda}_0$. The *reverse* protocol, used for Crooks, runs $x_1 \to x_0$ in the same time, starting in equilibrium at $\boldsymbol{\lambda}_1$.

Units and parameters. $m = b = \varepsilon = 1$; $k = 100$, $k_t = 10$, $\gamma = 1$ ($\zeta = m\gamma$), $\Delta t = 0.01$, $M = 10^5$ pulls per parameter set; $k_\mathrm{B}T \in \{0.25, 0.5, 1, 2\}$, $N \in \{5, 10, 20, 40\}$.

3 The heat bath: Langevin dynamics

Each mobile bead obeys

$$\mathrm{d}\boldsymbol{r}_i = \frac{\boldsymbol{p}_i}{m}\,\mathrm{d}t, \qquad \mathrm{d}\boldsymbol{p}_i = -\nabla_{\boldsymbol{r}_i} H(\boldsymbol{x};\boldsymbol{\lambda}(t))\,\mathrm{d}t - \gamma\,\boldsymbol{p}_i\,\mathrm{d}t + \sqrt{2\gamma m k_\mathrm{B}T}\,\mathrm{d}\boldsymbol{B}_i(t), \tag{2}$$

with independent 2D Wiener processes $\boldsymbol{B}_i$.

Why the noise strength must be $2\gamma m k_\mathrm{B}T$. For one free bead in 1D, $\mathrm{d}p = -\gamma p\,\mathrm{d}t + \sigma\,\mathrm{d}B$. Itô's lemma gives $\mathrm{d}(p^2) = 2p\,\mathrm{d}p + \sigma^2\mathrm{d}t$, hence $\frac{\mathrm{d}}{\mathrm{d}t}\langle p^2\rangle = -2\gamma\langle p^2\rangle + \sigma^2$ and $\langle p^2\rangle \to \sigma^2/2\gamma$. Equipartition ($\langle p^2\rangle = mk_\mathrm{B}T$) requires $\sigma^2 = 2\gamma m k_\mathrm{B}T$. With $\gamma = 0$ ("random forces only") $\langle p^2\rangle = \sigma^2 t$ grows without bound.

3.1 Stationary distribution: why the bath produces the canonical ensemble

Why we need this. Three things in this project rest on one fact: at a frozen trap position $\boldsymbol{\lambda}$, the Langevin dynamics (2) relaxes to the canonical distribution

$$\rho_\mathrm{eq}(\boldsymbol{x};\boldsymbol{\lambda}) = \frac{e^{-\beta H(\boldsymbol{x};\boldsymbol{\lambda})}}{Z(\boldsymbol{\lambda})}, \qquad Z(\boldsymbol{\lambda}) = \int e^{-\beta H(\boldsymbol{x};\boldsymbol{\lambda})}\,\mathrm{d}\boldsymbol{x}, \qquad \beta = \frac{1}{k_\mathrm{B}T}. \tag{3}$$

(i) The free energy $F = -k_\mathrm{B}T\ln Z$ is defined with respect to it. (ii) Every pull must start from it. (iii) The proof of the Jarzynski equality (Sec. 5) uses *only* the property that the dynamics at frozen $\boldsymbol{\lambda}$ leaves ρ_eq unchanged. This subsection shows where that property comes from.

Two descriptions of the same dynamics. Equation (2) describes one random trajectory. Averaging over the noise gives the evolution of the probability density $\rho(\boldsymbol{x}, t)$ of the state. The bridge between the two is the *generator* $\mathcal{L}$, defined by how the average of any observable $f(\boldsymbol{x})$ changes:

$$\frac{\mathrm{d}}{\mathrm{d}t}\langle f(\boldsymbol{x}_t)\rangle = \langle(\mathcal{L}f)(\boldsymbol{x}_t)\rangle\,. \tag{4}$$

It follows from Itô's lemma: $\mathrm{d}f = \nabla_{\boldsymbol{r}} f\cdot\mathrm{d}\boldsymbol{r} + \nabla_{\boldsymbol{p}} f\cdot\mathrm{d}\boldsymbol{p} + \frac{1}{2}(2\gamma m k_\mathrm{B}T)\Delta_{\boldsymbol{p}} f\,\mathrm{d}t$. The last term comes from $(\mathrm{d}B)^2 = \mathrm{d}t$; the noise term itself averages to zero. Inserting (2),

$$\mathcal{L}f = \underbrace{\frac{\boldsymbol{p}}{m}\cdot\nabla_{\boldsymbol{r}} f - \nabla_{\boldsymbol{r}} H\cdot\nabla_{\boldsymbol{p}} f}_{\mathcal{L}^\mathrm{ham} f\ \text{(Newton)}} + \gamma\underbrace{\big(-\boldsymbol{p}\cdot\nabla_{\boldsymbol{p}} f + mk_\mathrm{B}T\,\Delta_{\boldsymbol{p}} f\big)}_{\mathcal{L}^\mathrm{OU} f\ \text{(friction + noise)}}. \tag{5}$$

Since $\langle f(\boldsymbol{x}_t)\rangle = \int f\rho_t\,\mathrm{d}\boldsymbol{x}$, an integration by parts moves $\mathcal{L}$ from f onto ρ. The density obeys the Fokker–Planck (here: Kramers) equation $\partial_t\rho = \mathcal{L}^\dagger\rho$, with the adjoint operator

$$\mathcal{L}^\dagger\rho = \underbrace{-\frac{\boldsymbol{p}}{m}\cdot\nabla_{\boldsymbol{r}}\rho + \nabla_{\boldsymbol{r}} H\cdot\nabla_{\boldsymbol{p}}\rho}_{\mathcal{L}^{\mathrm{ham}\dagger}\rho} + \gamma\underbrace{\nabla_{\boldsymbol{p}}\cdot\big(\boldsymbol{p}\,\rho + mk_\mathrm{B}T\,\nabla_{\boldsymbol{p}}\rho\big)}_{\mathcal{L}^{\mathrm{OU}\dagger}\rho}. \tag{6}$$

A density is *stationary* if $\mathcal{L}^\dagger\rho = 0$. We check the two parts separately for $\rho = e^{-\beta H}/Z$.

Hamiltonian part: any function of the energy is stationary. Take any $\rho = \phi(H)$. Then $\nabla_{\boldsymbol{r}}\rho = \phi'\,\nabla_{\boldsymbol{r}}H$ and $\nabla_{\boldsymbol{p}}\rho = \phi'\,\boldsymbol{p}/m$, so

$$\mathcal{L}^{\mathrm{ham}\dagger}\rho = \phi'\Big(-\frac{\boldsymbol{p}}{m}\cdot\nabla_{\boldsymbol{r}}H + \nabla_{\boldsymbol{r}}H\cdot\frac{\boldsymbol{p}}{m}\Big) = 0.$$

This is energy conservation (Liouville). Newton's equations alone do not choose a temperature: $e^{-\beta H}$ is stationary for *every* β, and so is any other function of H.

Friction and noise select the temperature. The OU part is the divergence of a probability current in momentum space, $\boldsymbol{J}_{\boldsymbol{p}} = -(\boldsymbol{p}\rho + mk_{\mathrm{B}}T\nabla_{\boldsymbol{p}}\rho)$. For $\rho \propto e^{-\beta H}$, $\nabla_{\boldsymbol{p}}\rho = -\beta(\boldsymbol{p}/m)\rho$, so $mk_{\mathrm{B}}T\,\nabla_{\boldsymbol{p}}\rho = -\boldsymbol{p}\rho$ and the current vanishes *identically*: friction pushes probability towards $\boldsymbol{p} = 0$, and the noise spreads it out again at exactly the same rate. Had we used a noise strength σ^2 instead of $2\gamma mk_{\mathrm{B}}T$, the current would be $\boldsymbol{p}\rho\,(\sigma^2/2\gamma mk_{\mathrm{B}}T - 1) \neq 0$ unless $\sigma^2 = 2\gamma mk_{\mathrm{B}}T$. Only one temperature is stationary, the one fixed by the fluctuation–dissipation relation.

Uniqueness and relaxation. Because the noise acts on the momenta and the forces couple momenta to positions, the dynamics explores the whole phase space. ρ_{eq} is then the *unique* stationary density, and any initial density relaxes to it (hypocoercivity). This is what "the chain is thermalised by the bath" means mathematically. When $\boldsymbol{\lambda}$ moves, $\rho_{\mathrm{eq}}(\boldsymbol{\lambda}(t))$ is no longer stationary and the chain lags behind it; the work that this lag costs is the dissipated work.

3.2 Slowest relaxation time τ_1

What it is for. Whether a pull is "fast" or "slow" only makes sense relative to how quickly the chain can re-equilibrate. We measure every pulling time in units of τ_1, the relaxation time of the slowest collective motion of the chain. If $\tau \gg \tau_1$ the chain follows the trap almost in equilibrium (little dissipation). If $\tau \lesssim \tau_1$ the chain lags behind and dissipates a lot. τ_1 sets the scale; it does not enter any of the measured quantities.

Step 1: overdamped description at the chain scale. Momenta relax on the time $1/\gamma = 1$, much shorter than the collective times we are after (tens to hundreds of time units). On those scales inertia is negligible, and each bead obeys the overdamped (Brownian) equation $\zeta\,\dot{\boldsymbol{r}}_i = \boldsymbol{F}_i + \text{noise}$, with friction coefficient $\zeta = m\gamma$.

Step 2: replace the chain by linear springs. For $b = 0$ the bonds are already linear springs of stiffness $\kappa = k$. For $b = 1$ the bonds hardly stretch, but a *segment* of the chain behaves as a spring of entropic origin. By the central limit theorem, the end-to-end vector of n freely jointed bonds is Gaussian, with each Cartesian component having variance $n\langle u^2\rangle/2$ in 2D. Its free energy is therefore $F(\boldsymbol{R}) = -k_{\mathrm{B}}T\ln P(\boldsymbol{R}) = k_{\mathrm{B}}T|\boldsymbol{R}|^2/(n\langle u^2\rangle) + \text{const}$: a harmonic spring of stiffness κ/n with

$$\kappa = \frac{2k_{\mathrm{B}}T}{\langle u^2\rangle} \qquad \text{per bond (2D entropic spring).} \tag{7}$$

This is the Rouse model: each bond is replaced by an entropic spring. For $k_{\mathrm{B}}T = 1$, $\kappa \approx 1.94$, about $50\times$ softer than the bond itself ($k = 100$).

Step 3: normal modes. With linear springs the overdamped equations read $\zeta\,\dot{\boldsymbol{r}} = -K\boldsymbol{r} + (\text{trap force at fixed } \boldsymbol{\lambda}) + \text{noise}$, separately for x and y. K is the $N \times N$ stiffness matrix: $K_{ii} = 2\kappa$ for $i < N$, $K_{NN} = \kappa + k_t$ (the last bead has one bond and the trap), $K_{i,i\pm1} = -\kappa$. The anchored bead 0 is fixed and does not appear. Writing the deviation from the mean shape in the eigenvectors of K, each mode p relaxes independently as $e^{-\lambda_p t/\zeta}$. The slowest one defines

$$\tau_1 = \frac{\zeta}{\lambda_{\min}(K)}, \tag{8}$$

which is what the code computes (`params.tau1`).

Step 4: closed-form estimate. Since $k_t = 10 \gg \kappa \approx 2$, the trap holds bead N almost fixed, so the chain is effectively clamped at both ends. The modes of a discrete string fixed at both ends are $\sin(p\pi i/N)$, with eigenvalues $\lambda_p = 4\kappa \sin^2(p\pi/2N) \simeq \kappa(p\pi/N)^2$. The slowest is the half-wave $p = 1$:

$$\tau_1 \simeq \frac{\zeta}{\kappa}\Big(\frac{N}{\pi}\Big)^2 = \frac{\zeta N^2 \langle u^2 \rangle}{2\pi^2 k_\mathrm{B} T} \qquad (b = 1). \tag{9}$$

For $N = 10$ this gives 5.22, against the exact eigenvalue 5.46 (the trap is not infinitely stiff). For $N = 40$ it gives 83.5, against 84.3.

What the formula says. $\tau_1 \propto N^2$ (Rouse scaling: a disturbance diffuses along the chain with a diffusion constant κ/ζ per bead^2) and $\tau_1 \propto \zeta$. For the entropic chain $\tau_1 \propto 1/T$, because the entropic spring is stiffer at high temperature; for the Gaussian chain ($\kappa = k$) τ_1 does not depend on T. Values: $\tau_1 = 1.45, 5.46, 21.3, 84.3$ for $N = 5, 10, 20, 40$ at $k_\mathrm{B}T = 1$; 20.8, 10.6, 5.46, 2.91 for $k_\mathrm{B}T = 0.25, 0.5, 1, 2$ at $N = 10$; 0.26 for the $b = 0$ chain with $N = 10$.

Correction to the first version: it assumed a *free* far end, whose slowest mode is the quarter wave $\sin(\pi i/2N)$. That overestimates τ_1 by a factor 4.

4 Equilibrium free energy

Definition. With the canonical distribution (3), the free energy at fixed $\boldsymbol{\lambda}$ is $F(\boldsymbol{\lambda}) = -k_\mathrm{B}T \ln Z(\boldsymbol{\lambda})$, and

$$\Delta F = F(\boldsymbol{\lambda}_1) - F(\boldsymbol{\lambda}_0) = -k_\mathrm{B}T \ln \frac{Z(\boldsymbol{\lambda}_1)}{Z(\boldsymbol{\lambda}_0)}. \tag{10}$$

Only *ratios* of partition functions are physical here, so constant factors in Z (the momentum integral, normalisation of $\mathrm{d}\boldsymbol{x}$) cancel. The kinetic energy does not depend on $\boldsymbol{\lambda}$, so ΔF is a purely configurational quantity.

ΔF is the reversible work. Differentiate F with respect to $\boldsymbol{\lambda}$:

$$\frac{\partial F}{\partial \boldsymbol{\lambda}} = -k_\mathrm{B}T \,\frac{1}{Z}\frac{\partial Z}{\partial \boldsymbol{\lambda}} = -k_\mathrm{B}T \,\frac{1}{Z}\int \Big(-\beta \frac{\partial H}{\partial \boldsymbol{\lambda}}\Big) e^{-\beta H}\mathrm{d}\boldsymbol{x} = \Big\langle \frac{\partial H}{\partial \boldsymbol{\lambda}} \Big\rangle_{\boldsymbol{\lambda}}. \tag{11}$$

The derivative of the free energy is the equilibrium average of the generalised force. For the trap this is $\partial_{\boldsymbol{\lambda}} H = -k_t(\boldsymbol{r}_N - \boldsymbol{\lambda})$, i.e. minus the mean force the trap exerts on the bead. If the trap is moved infinitely slowly, the chain stays in equilibrium at every $\boldsymbol{\lambda}$, so the work is $W_\mathrm{rev} = \int \langle \partial_{\boldsymbol{\lambda}} H \rangle_{\boldsymbol{\lambda}} \cdot \mathrm{d}\boldsymbol{\lambda} = \int \partial_{\boldsymbol{\lambda}} F \cdot \mathrm{d}\boldsymbol{\lambda} = \Delta F$. This is also the basis of *thermodynamic integration*: measure the mean force at a grid of fixed $\boldsymbol{\lambda}$ and integrate. It is used as a check in the code.

Energy and entropy. $\Delta F = \Delta U - T\Delta S$ with $\Delta S = -\partial \Delta F/\partial T$. For the stiff-bond chain $\Delta F \approx T\,\phi(\boldsymbol{\lambda}_0, \boldsymbol{\lambda}_1)$ (Sec. 9.2), so $\Delta U \approx 0$ and $\Delta F \approx -T\Delta S$: the work goes into lowering the entropy of the chain, and the same amount is released to the bath as heat. For the Gaussian chain ΔF is independent of T and is entirely stored as spring energy.

5 Work, heat and the Jarzynski equality

5.1 Work and heat along one trajectory

Following Sekimoto, the change of energy along a trajectory splits into a part due to the control and a part exchanged with the bath:

$$\mathrm{d}H = \underbrace{\frac{\partial H}{\partial \boldsymbol{\lambda}} \cdot \mathrm{d}\boldsymbol{\lambda}}_{\mathrm{d}W} + \underbrace{\nabla_{\boldsymbol{x}} H \circ \mathrm{d}\boldsymbol{x}}_{\mathrm{d}Q}, \qquad W = \int_0^\tau \frac{\partial H}{\partial \boldsymbol{\lambda}} \cdot \dot{\boldsymbol{\lambda}}\, \mathrm{d}t = -k_t \int_0^\tau (\boldsymbol{r}_N - \boldsymbol{\lambda}) \cdot \dot{\boldsymbol{\lambda}}\, \mathrm{d}t. \tag{12}$$

W is the work done by the puller and $Q = \Delta H - W$ the heat received from the bath.

Jarzynski equality (1997)

If (i) the system starts in equilibrium, $\boldsymbol{x}(0) \sim \rho_{\mathrm{eq}}(\cdot; \boldsymbol{\lambda}_0)$, and (ii) at every frozen $\boldsymbol{\lambda}$ the dynamics leaves $\rho_{\mathrm{eq}}(\cdot; \boldsymbol{\lambda})$ invariant, then for *any* protocol $\boldsymbol{\lambda}(t)$ and *any* duration τ,

$$\left\langle e^{-\beta W} \right\rangle = e^{-\beta \Delta F}. \tag{13}$$

Neither detailed balance nor equilibrium at the end ($t = \tau$) is required.

5.2 Proof 1: isolated Hamiltonian system (the original argument)

Suppose the system is first equilibrated at $\boldsymbol{\lambda}_0$, then decoupled from the bath and driven by Hamilton's equations. With no bath there is no heat, so $W = H(\boldsymbol{x}_\tau; \boldsymbol{\lambda}_1) - H(\boldsymbol{x}_0; \boldsymbol{\lambda}_0)$, where $\boldsymbol{x}_\tau = \Phi(\boldsymbol{x}_0)$ is the deterministic image of the initial point. Then

$$\begin{aligned} \left\langle e^{-\beta W} \right\rangle &= \int \mathrm{d}\boldsymbol{x}_0\, \frac{e^{-\beta H(\boldsymbol{x}_0;\boldsymbol{\lambda}_0)}}{Z_0}\, e^{-\beta[H(\Phi(\boldsymbol{x}_0);\boldsymbol{\lambda}_1) - H(\boldsymbol{x}_0;\boldsymbol{\lambda}_0)]} = \frac{1}{Z_0} \int \mathrm{d}\boldsymbol{x}_0\, e^{-\beta H(\Phi(\boldsymbol{x}_0);\boldsymbol{\lambda}_1)} \\ &= \frac{1}{Z_0} \int \mathrm{d}\boldsymbol{x}_\tau\, e^{-\beta H(\boldsymbol{x}_\tau;\boldsymbol{\lambda}_1)} = \frac{Z_1}{Z_0} = e^{-\beta \Delta F}. \end{aligned} \tag{14}$$

The first step cancels the initial Boltzmann factor against the $e^{+\beta H(\boldsymbol{x}_0;\boldsymbol{\lambda}_0)}$ inside $e^{-\beta W}$. The change of variables $\boldsymbol{x}_0 \to \boldsymbol{x}_\tau$ has unit Jacobian by Liouville's theorem. The whole proof is "the initial Boltzmann weight times $e^{-\beta W}$ equals the final Boltzmann weight, and phase-space volume is conserved".

5.3 Proof 2: discrete-time Markov dynamics (what the code implements)

Split the protocol into steps $\boldsymbol{\lambda}_0, \boldsymbol{\lambda}_1, \ldots, \boldsymbol{\lambda}_K$. Each step does:

1. *switch* $\boldsymbol{\lambda}_k \to \boldsymbol{\lambda}_{k+1}$ at frozen $\boldsymbol{x}_k$, costing work $\delta W_k = H(\boldsymbol{x}_k; \boldsymbol{\lambda}_{k+1}) - H(\boldsymbol{x}_k; \boldsymbol{\lambda}_k)$;
2. *relax*: draw $\boldsymbol{x}_{k+1}$ from a Markov kernel $P_{k+1}(\boldsymbol{x}_k \to \boldsymbol{x}_{k+1})$ that preserves $\rho_{\mathrm{eq}}(\cdot; \boldsymbol{\lambda}_{k+1})$, i.e. $\int \mathrm{d}\boldsymbol{x}\, e^{-\beta H(\boldsymbol{x};\boldsymbol{\lambda}_{k+1})} P_{k+1}(\boldsymbol{x} \to \boldsymbol{y}) = e^{-\beta H(\boldsymbol{y};\boldsymbol{\lambda}_{k+1})}$.

The total work is $W = \sum_{k=0}^{K-1} \delta W_k$. Write the average as an integral over the whole path:

$$\left\langle e^{-\beta W} \right\rangle = \int \mathrm{d}\boldsymbol{x}_0 \cdots \mathrm{d}\boldsymbol{x}_K\, \frac{e^{-\beta H(\boldsymbol{x}_0;\boldsymbol{\lambda}_0)}}{Z_0} \prod_{k=0}^{K-1} e^{-\beta[H(\boldsymbol{x}_k;\boldsymbol{\lambda}_{k+1}) - H(\boldsymbol{x}_k;\boldsymbol{\lambda}_k)]}\, P_{k+1}(\boldsymbol{x}_k \to \boldsymbol{x}_{k+1}). \tag{15}$$

Integrate over $\boldsymbol{x}_0$ first. The factors containing $\boldsymbol{x}_0$ are $e^{-\beta H(\boldsymbol{x}_0;\boldsymbol{\lambda}_0)} e^{-\beta[H(\boldsymbol{x}_0;\boldsymbol{\lambda}_1) - H(\boldsymbol{x}_0;\boldsymbol{\lambda}_0)]} P_1(\boldsymbol{x}_0 \to \boldsymbol{x}_1) = e^{-\beta H(\boldsymbol{x}_0;\boldsymbol{\lambda}_1)} P_1(\boldsymbol{x}_0 \to \boldsymbol{x}_1)$. By invariance, $\int \mathrm{d}\boldsymbol{x}_0\, e^{-\beta H(\boldsymbol{x}_0;\boldsymbol{\lambda}_1)} P_1(\boldsymbol{x}_0 \to \boldsymbol{x}_1) = e^{-\beta H(\boldsymbol{x}_1;\boldsymbol{\lambda}_1)}$. The remaining integrand has exactly the same form with $0 \to 1$. Repeating K times leaves

$$\left\langle e^{-\beta W} \right\rangle = \frac{1}{Z_0} \int \mathrm{d}\boldsymbol{x}_K\, e^{-\beta H(\boldsymbol{x}_K;\boldsymbol{\lambda}_K)} = \frac{Z_K}{Z_0} = e^{-\beta \Delta F}. \tag{16}$$

In words: the weight $e^{-\beta\delta W_k}$ converts the Boltzmann factor of $\boldsymbol{\lambda}_k$ into that of $\boldsymbol{\lambda}_{k+1}$, and the relaxation step leaves that Boltzmann factor unchanged. *Consequences for the code:* the work must be accumulated as the energy jump at the switch, evaluated at the current positions, and the identity holds to the accuracy with which one integrator step preserves ρ_{eq} (Sec. 10).

5.4 Proof 3: continuous-time Langevin dynamics (Feynman–Kac)

Define the work-weighted density $g(\boldsymbol{x},t) = \left\langle \delta(\boldsymbol{x}(t)-\boldsymbol{x})\, e^{-\beta W_t}\right\rangle$, with $W_t = \int_0^t \dot{\boldsymbol{\lambda}}\cdot\partial_{\boldsymbol{\lambda}} H \,\mathrm{d}s$. Because W_t grows at rate $\dot{\boldsymbol{\lambda}}\cdot\partial_{\boldsymbol{\lambda}} H(\boldsymbol{x}(t);\boldsymbol{\lambda}(t))$, the Feynman–Kac formula gives

$$\partial_t g = \mathcal{L}^{\dagger}_{\boldsymbol{\lambda}(t)} g - \beta\, \dot{\boldsymbol{\lambda}}\cdot\partial_{\boldsymbol{\lambda}} H(\boldsymbol{x};\boldsymbol{\lambda}(t))\, g, \qquad g(\boldsymbol{x},0) = \rho_{\mathrm{eq}}(\boldsymbol{x};\boldsymbol{\lambda}_0). \tag{17}$$

Try $g = e^{-\beta H(\boldsymbol{x};\boldsymbol{\lambda}(t))}/Z_0$. The left side is $-\beta\,\dot{\boldsymbol{\lambda}}\cdot\partial_{\boldsymbol{\lambda}} H\, g$. On the right, $\mathcal{L}^{\dagger}_{\boldsymbol{\lambda}(t)} g = 0$ because g is proportional to the instantaneous Boltzmann factor, and the second term equals the left side. The initial condition matches, so by uniqueness this is the solution, and $\left\langle e^{-\beta W_t}\right\rangle = \int g\,\mathrm{d}\boldsymbol{x} = Z(\boldsymbol{\lambda}(t))/Z_0$.

6 From the equality to the free energy

The formula. Taking the logarithm of (13),

$$\Delta F = -k_{\mathrm{B}}T \ln\left\langle e^{-\beta W}\right\rangle. \tag{18}$$

The right side is a property of a nonequilibrium process: the pull is arbitrarily fast and the chain at $t=\tau$ is far from equilibrium. The left side is a property of two equilibrium states. The equality therefore turns irreversible work measurements into an *equilibrium* free-energy difference.

What is and is not determined.

- Only differences of F are accessible: Z_0 appears as a normalisation and cancels, so $F(\boldsymbol{\lambda}_0)$ serves as the reference.
- Because the proofs hold at every intermediate time, the same pulls give the whole *profile*: $F(\boldsymbol{\lambda}(t)) - F(\boldsymbol{\lambda}_0) = -k_{\mathrm{B}}T\ln\left\langle e^{-\beta W_t}\right\rangle$ for every $t \le \tau$ (Hummer & Szabo 2001).
- In the quasi-static limit the work stops fluctuating, $W \to W_{\mathrm{rev}} = \Delta F$ (Sec. 4), and (18) becomes trivial. The equality is non-trivial precisely when W fluctuates.

Second law. The exponential is convex, so Jensen's inequality gives $e^{-\beta\langle W\rangle} \le \left\langle e^{-\beta W}\right\rangle = e^{-\beta\Delta F}$, i.e.

$$\langle W\rangle \ge \Delta F, \qquad W_{\mathrm{diss}} \equiv \langle W\rangle - \Delta F \ge 0. \tag{19}$$

The second law holds *on average*; individual pulls can have $W < \Delta F$. They *must* exist whenever $W_{\mathrm{diss}} > 0$, otherwise $\left\langle e^{-\beta W}\right\rangle < e^{-\beta\Delta F}$.

How rare are "violations"? For any $\zeta > 0$,

$$\Pr(W < \Delta F - \zeta) = \int_{-\infty}^{\Delta F-\zeta} P(W)\,\mathrm{d}W \le \int P(W)\, e^{\beta(\Delta F-\zeta-W)}\mathrm{d}W = e^{-\beta\zeta}, \tag{20}$$

since the exponential factor is ≥ 1 on the integration range. Violations by more than a few $k_{\mathrm{B}}T$ are exponentially rare. They are also exactly the pulls that dominate (18), which is the root of all the sampling difficulty (Sec. 8).

7 Crooks fluctuation theorem

Crooks fluctuation theorem (1999)

Let $P_F(W)$ be the work distribution of the forward protocol ($\boldsymbol{\lambda}_0 \to \boldsymbol{\lambda}_1$, started in equilibrium at $\boldsymbol{\lambda}_0$) and $P_R(W)$ that of the time-reversed protocol ($\boldsymbol{\lambda}_1 \to \boldsymbol{\lambda}_0$, started in equilibrium at $\boldsymbol{\lambda}_1$). Then

$$\frac{P_F(W)}{P_R(-W)} = e^{\beta(W-\Delta F)}. \tag{21}$$

Derivation (discrete time). Assume each relaxation kernel satisfies detailed balance, $P_k(\boldsymbol{x} \to \boldsymbol{y})\, e^{-\beta H(\boldsymbol{x};\boldsymbol{\lambda}_k)} = P_k(\boldsymbol{y} \to \boldsymbol{x})\, e^{-\beta H(\boldsymbol{y};\boldsymbol{\lambda}_k)}$. For underdamped dynamics the same holds with momenta reversed in the backward path. Compare the probability of a forward path $\boldsymbol{x}_0 \to \cdots \to \boldsymbol{x}_K$ with the probability that the reverse protocol produces the reversed path $\boldsymbol{x}_K \to \cdots \to \boldsymbol{x}_0$:

$$\frac{\mathcal{P}_F[\boldsymbol{x}]}{\mathcal{P}_R[\tilde{\boldsymbol{x}}]} = \frac{\rho_{\mathrm{eq}}(\boldsymbol{x}_0;\boldsymbol{\lambda}_0)}{\rho_{\mathrm{eq}}(\boldsymbol{x}_K;\boldsymbol{\lambda}_K)} \prod_{k=0}^{K-1} \frac{P_{k+1}(\boldsymbol{x}_k \to \boldsymbol{x}_{k+1})}{P_{k+1}(\boldsymbol{x}_{k+1} \to \boldsymbol{x}_k)} = \frac{e^{-\beta H(\boldsymbol{x}_0;\boldsymbol{\lambda}_0)} Z_K}{e^{-\beta H(\boldsymbol{x}_K;\boldsymbol{\lambda}_K)} Z_0} \prod_k e^{-\beta[H(\boldsymbol{x}_{k+1};\boldsymbol{\lambda}_{k+1}) - H(\boldsymbol{x}_k;\boldsymbol{\lambda}_{k+1})]}. \tag{22}$$

The product is $e^{-\beta Q}$, where Q is the heat (the energy change during the relaxation steps). The prefactor is $e^{\beta\Delta H} e^{-\beta\Delta F}$, so the ratio is $e^{\beta(\Delta H - Q) - \beta\Delta F} = e^{\beta(W-\Delta F)}$. The reversed path has work $-W$ under the reverse protocol. Summing over all paths with forward work W gives (21).

Consequences.

1. *Crossing.* $P_F(W)$ and $P_R(-W)$ intersect exactly at $W = \Delta F$. This gives a graphical estimate of ΔF.

2. *Linear log-ratio.* $\ln[P_F(W)/P_R(-W)] = \beta W - \beta\Delta F$, a straight line of slope β, independent of the protocol. This is the test used in the report.

3. *Jarzynski follows.* Write (21) as $P_F(W) e^{-\beta W} = e^{-\beta\Delta F} P_R(-W)$ and integrate over W: $\left\langle e^{-\beta W} \right\rangle_F = e^{-\beta\Delta F} \int P_R(-W)\, \mathrm{d}W = e^{-\beta\Delta F}$.

4. *Where the exponential average comes from.* Rearranging,

$$\frac{e^{-\beta W} P_F(W)}{\left\langle e^{-\beta W} \right\rangle_F} = P_R(-W). \tag{23}$$

   The share of $\left\langle e^{-\beta W} \right\rangle_F$ contributed by forward works near W is exactly the reverse work distribution, mirrored. The forward average is dominated by works typical of the *reverse* process, near $W \approx -\langle W \rangle_R = \Delta F - W^R_{\mathrm{diss}}$. The report's plots of $e^{-\beta W}$ show this directly.

5. *Gaussian case.* If both distributions are Gaussian, (21) forces equal variances σ^2 and $\langle W \rangle_F - \Delta F = \langle W \rangle_R + \Delta F = \beta\sigma^2/2$: forward and reverse dissipation are equal. In general they are not (the report finds $W^R_{\mathrm{diss}} < W^F_{\mathrm{diss}}$).

8 Estimators of ΔF

We have M i.i.d. forward works $W_1, \ldots, W_M$ and, for the two-sided estimators, M_R reverse works $W^R_1, \ldots, W^R_{M_R}$.

8.1 One-sided Jarzynski estimator

Replace the average in (18) by the sample mean:

$$\widehat{\Delta F}_J = -k_{\mathrm{B}}T \ln \bar{Y}, \qquad \bar{Y} = \frac{1}{M}\sum_{m=1}^{M} e^{-\beta W_m}. \tag{24}$$

Consistency. By the law of large numbers $\bar{Y} \to \langle e^{-\beta W}\rangle = e^{-\beta \Delta F}$, and ln is continuous, so $\widehat{\Delta F}_J \to \Delta F$ as $M \to \infty$.

Bias and variance (delta method). Write $\bar{Y} = \mu(1+\epsilon)$ with $\mu = e^{-\beta\Delta F}$, $\langle \epsilon \rangle = 0$ and $\langle \epsilon^2 \rangle = v/M$, where $v = \mathrm{Var}(e^{-\beta W})/\mu^2$ is the relative variance of a single weight. Expanding $\ln(1+\epsilon) = \epsilon - \epsilon^2/2 + \dots$,

$$\left\langle \widehat{\Delta F}_J \right\rangle - \Delta F \simeq \frac{k_{\mathrm{B}}T}{2}\frac{v}{M} > 0, \qquad \mathrm{Var}\,\widehat{\Delta F}_J \simeq (k_{\mathrm{B}}T)^2 \frac{v}{M}, \qquad v = \frac{\langle e^{-2\beta W}\rangle}{\langle e^{-\beta W}\rangle^2} - 1. \tag{25}$$

The bias is always *upward* (Jensen again: ln is concave). For Gaussian work, $\langle e^{-2\beta W}\rangle / \langle e^{-\beta W}\rangle^2 = e^{\beta^2\sigma^2} = e^{2\beta W_{\mathrm{diss}}}$, so

$$\text{bias} \simeq \frac{k_{\mathrm{B}}T}{2M}\left(e^{2\beta W_{\mathrm{diss}}} - 1\right), \qquad \text{s.d.} \simeq k_{\mathrm{B}}T\sqrt{\frac{e^{2\beta W_{\mathrm{diss}}} - 1}{M}}. \tag{26}$$

These asymptotic formulas are valid only once $M \gg e^{2\beta W_{\mathrm{diss}}}$, i.e. when the error is already smaller than $k_{\mathrm{B}}T$.

The intermediate regime (rare events). By (23), the pulls that matter are those typical of the reverse process. Under P_F they are a tail event of probability $\sim e^{-\beta W^R_{\mathrm{diss}}}$, so one needs

$$M \gtrsim e^{\beta W^R_{\mathrm{diss}}} \tag{27}$$

forward pulls just to *see* them (Jarzynski 2006). With fewer pulls, $\widehat{\Delta F}_J$ sits between ΔF and $\langle W \rangle$, typically too high. A single estimate can still fall below ΔF if one lucky pull lands deep in the tail. For Gaussian work $W^R_{\mathrm{diss}} = W_{\mathrm{diss}}$, and (27) is the criterion used in the report.

Error bars. The bootstrap resamples only the pulls that were observed and cannot represent tail pulls that were never drawn, so it underestimates the error when few pulls dominate. Independent repetitions, or the block-convergence curve (estimate vs. number of pulls used), are more honest.

8.2 Cumulant (Gaussian) estimators

The cumulant generating function of W is $\ln\langle e^{sW}\rangle = \sum_{n\geq 1} \kappa_n s^n/n!$. At $s = -\beta$,

$$\Delta F = -k_{\mathrm{B}}T \ln\left\langle e^{-\beta W}\right\rangle = \kappa_1 - \frac{\beta}{2}\kappa_2 + \frac{\beta^2}{6}\kappa_3 - \dots \tag{28}$$

Truncating after κ_2 gives $\widehat{\Delta F}_2 = \overline{W} - \beta s_W^2/2$, exact for Gaussian work (e.g. the $b = 0$ chain) and low-variance, but biased otherwise. With positive skewness ($\kappa_3 > 0$, a long right tail) the truncation *underestimates* ΔF. Near equilibrium (linear response) $P(W)$ is Gaussian and $W_{\mathrm{diss}} = \beta\sigma_W^2/2$, the work fluctuation–dissipation relation.

8.3 Effective sample size

$\widehat{\Delta F}_J$ is a weighted average with weights $w_m = e^{-\beta W_m}$. The Kish effective sample size $M_{\mathrm{eff}} = (\sum_m w_m)^2 / \sum_m w_m^2$ is the number of equally weighted samples that would give the same variance for a weighted mean. It equals M when all weights are equal and 1 when one pull dominates. Using (25), $M_{\mathrm{eff}} \approx M/(1+v)$ for large M.

8.4 Reverse Jarzynski estimator

Applying (13) to the reverse protocol gives $\langle e^{-\beta W_R}\rangle = e^{+\beta\Delta F}$, so

$$\widehat{\Delta F}_R = +k_{\mathrm{B}}T \ln\Big(\frac{1}{M_R}\sum_m e^{-\beta W_m^R}\Big). \tag{29}$$

It needs $M_R \gtrsim e^{\beta W^F_{\mathrm{diss}}}$ and is biased *downward*.

8.5 Bennett acceptance ratio (BAR)

Multiply Crooks (21) by an arbitrary function $f(W)$ and integrate:

$$\Big\langle f(W)\, e^{-\beta W}\Big\rangle_F = e^{-\beta\Delta F}\,\langle f(-W)\rangle_R \quad\Longrightarrow\quad e^{-\beta\Delta F} = \frac{\langle f(W)e^{-\beta W}\rangle_F}{\langle f(-W)\rangle_R} \quad \text{for every } f. \tag{30}$$

$f = 1$ recovers forward Jarzynski and $f = e^{\beta W}$ recovers reverse Jarzynski. Bennett (1976) showed that the choice minimising the asymptotic variance is

$$f(W) = \frac{e^{\beta W}}{1 + \frac{M}{M_R}e^{\beta(W-\Delta F)}}, \tag{31}$$

which depends on the unknown ΔF. Substituting it and simplifying with the Fermi function $\mathrm{f}(x) = 1/(1+e^x)$ gives the self-consistent BAR equation

$$\sum_{m=1}^{M} \mathrm{f}\Big(\beta(W_m - \widehat{\Delta F}) + \ln\tfrac{M}{M_R}\Big) = \sum_{m=1}^{M_R} \mathrm{f}\Big(\beta(W_m^R + \widehat{\Delta F}) - \ln\tfrac{M}{M_R}\Big), \tag{32}$$

whose left side increases and right side decreases with $\widehat{\Delta F}$, so the root is unique (solved by bracketing). BAR is also the maximum-likelihood estimator of ΔF given both samples (Shirts et al. 2003). It weights each pull by how close it is to the *overlap* region between $P_F(W)$ and $P_R(-W)$, around $W \approx \Delta F$. It works as long as the two distributions overlap, which requires far fewer pulls than either one-sided estimator, since each one-sided estimator must reach deep into the other distribution's typical region.

9 Exact free-energy references

In bond variables $\boldsymbol{u}_i = \boldsymbol{r}_i - \boldsymbol{r}_{i-1}$ (unit Jacobian, $\boldsymbol{r}_N = \sum \boldsymbol{u}_i$) the bonds are independent, and

$$Z(\boldsymbol{\lambda}) \propto \big(g^{*N} * w\big)(\boldsymbol{\lambda}), \qquad g(\boldsymbol{u}) = e^{-\beta U_b(|\boldsymbol{u}|)}, \quad w(\boldsymbol{s}) = e^{-\beta k_t|\boldsymbol{s}|^2/2}\,: \tag{33}$$

the trap acts as an $(N{+}1)$-th bond with zero rest length.

9.1 Gaussian chain ($b = 0$): closed form

All factors are Gaussian, and variances add (springs in series):

$$\Delta F = \tfrac{1}{2}k_{\mathrm{eff}}\big(|\boldsymbol{\lambda}_1|^2 - |\boldsymbol{\lambda}_0|^2\big), \qquad 1/k_{\mathrm{eff}} = N/k + 1/k_t. \tag{34}$$

9.2 Stiff-bond chain ($b = 1$): numerically exact

Keeping only the second moment of each bond gives the entropic-spring approximation $\Delta F \approx k_{\mathrm{B}}T(|\boldsymbol{\lambda}_1|^2 - |\boldsymbol{\lambda}_0|^2)/(N\langle u^2\rangle + 2k_{\mathrm{B}}T/k_t) \propto T$. The exact value iterates the radial 2D convolution in log space, $\ln h_{n+1}(R) = \ln\int r\,\mathrm{d}r\,\mathrm{d}\theta\, g(r)h_n\big(\sqrt{R^2+r^2-2Rr\cos\theta}\big)$, with log-sum-exp quadrature. The equivalent Fourier (Hankel) route cancels catastrophically once $Z(\boldsymbol{\lambda}_1)/Z(\boldsymbol{\lambda}_0) \lesssim 10^{-4}$. The method reproduces the closed form to 10^{-4} and agrees with the Hankel route wherever that route is reliable. Values are tabulated in the report.

10 Numerical scheme

10.1 Integrator and work bookkeeping

Integrator (BAOAB). Kick (B), drift (A) and Ornstein–Uhlenbeck (O) sub-steps, each solved exactly, applied in the order B–A–O–A–B: B : $\boldsymbol{p}$ += $\frac{\Delta t}{2}\boldsymbol{F}$; A : $\boldsymbol{r}$ += $\frac{\Delta t}{2m}\boldsymbol{p}$; O : $\boldsymbol{p} \leftarrow e^{-\gamma\Delta t}\boldsymbol{p} + \sqrt{(1-e^{-2\gamma\Delta t})mk_{\mathrm{B}}T}\,\boldsymbol{\xi}$. The O step is exact for any Δt. BAOAB has the smallest configurational bias among standard splittings (Leimkuhler & Matthews 2013).

Work bookkeeping. Every step: (1) W += $\frac{k_t}{2}[(x_N-\lambda_{k+1})^2-(x_N-\lambda_k)^2]$ at the current x_N; (2) one BAOAB step at $\boldsymbol{\lambda}_{k+1}$. This is the switch/relax scheme of Proof 2, so the only source of error in (13) is the $O(\Delta t^2)$ invariant-measure error of BAOAB.

10.2 Exact initial ensemble: tilted rejection sampling

Goal. Every pull must start from the equilibrium distribution $\rho_{\mathrm{eq}}(\cdot;\boldsymbol{\lambda})$ at the initial trap position: $\boldsymbol{\lambda}_0$ (chain at 20% extension) for forward pulls, $\boldsymbol{\lambda}_1$ (70%) for reverse pulls. One option is to run the dynamics at fixed $\boldsymbol{\lambda}$ long enough to forget the starting point (several τ_1 per chain). The phantom chain allows something better: drawing *exact*, independent equilibrium configurations directly. Momenta are simply Maxwellian, independent of positions, so only the positions are non-trivial.

Step 1: what the target looks like in bond variables. In bond vectors $\boldsymbol{u}_i = \boldsymbol{r}_i - \boldsymbol{r}_{i-1}$ the positional Boltzmann weight is

$$\rho(\boldsymbol{u}_1,\dots,\boldsymbol{u}_N) \propto \underbrace{\prod_{i=1}^{N} g(\boldsymbol{u}_i)}_{\text{independent bonds}} \times \underbrace{w(\boldsymbol{\lambda}-\boldsymbol{R})}_{\text{trap couples them}}, \qquad \boldsymbol{R}=\sum_i \boldsymbol{u}_i, \tag{35}$$

with $g(\boldsymbol{u}) = e^{-\beta U_b(|\boldsymbol{u}|)}$ and $w(\boldsymbol{s}) = e^{-\beta k_t|\boldsymbol{s}|^2/2}$. Without the trap the bonds would be independent and trivial to sample; the trap only cares about the end point $\boldsymbol{R}$.

Step 2: plain rejection sampling. Rejection sampling draws from an easy proposal q and keeps each draw with probability proportional to ρ/q, which requires ρ/q to be bounded. Take $q = \prod_i g(\boldsymbol{u}_i)$, i.e. a free chain with no trap. Then $\rho/q \propto w(\boldsymbol{\lambda}-\boldsymbol{R}) \le 1$: draw a free chain and keep it with probability $w(\boldsymbol{\lambda}-\boldsymbol{R})$, the Boltzmann factor of the trap spring. The kept chains are distributed *exactly* as ρ. The method fails when $\boldsymbol{\lambda}$ is far from where free chains end. A free chain of $N = 40$ unit bonds has $|\boldsymbol{R}| \sim \sqrt{40} \approx 6$, while the reverse pulls start at $\boldsymbol{\lambda}_1 = 28$: the acceptance rate is $\sim 10^{-9}$.

Step 3: propose chains that are already stretched (the tilt). Replace the free-chain proposal by a chain whose end is pulled by a constant force f along x, as if a weight hung on it:

$$q_f(\boldsymbol{u}_1,\dots,\boldsymbol{u}_N) \propto \prod_i g(\boldsymbol{u}_i)\, e^{\beta f u_{i,x}} = \Big(\prod_i g(\boldsymbol{u}_i)\Big) e^{\beta f R_x}. \tag{36}$$

The bonds are *still independent*, so q_f remains easy to sample. In polar coordinates $\boldsymbol{u} = r(\cos\theta,\sin\theta)$ one bond has density $\propto r\,g(r)\,e^{\beta f r\cos\theta}$:

- the length r has marginal density $\propto r\,g(r)\,I_0(\beta f r)$ (integrating $e^{a\cos\theta}$ over θ gives $2\pi I_0(a)$), drawn by inverse CDF on a fine grid;
- given r, the angle has density $\propto e^{\beta f r\cos\theta}$, a von Mises distribution with concentration $\beta f r$ (a standard random-number routine).

Step 4: undo the tilt exactly. The ratio of target to proposal is $\rho/q_f \propto w(\boldsymbol{\lambda}-\boldsymbol{R})\, e^{-\beta f R_x}$. Completing the square,

$$-\tfrac{\beta k_t}{2}|\boldsymbol{R}-\boldsymbol{\lambda}|^2 - \beta f R_x = -\tfrac{\beta k_t}{2}|\boldsymbol{R}-\boldsymbol{\lambda}'|^2 + \text{const}, \qquad \boldsymbol{\lambda}' = \boldsymbol{\lambda} - \frac{f}{k_t}\hat{\boldsymbol{x}}. \tag{37}$$

So we accept with probability $a(\boldsymbol{R}) = e^{-\beta k_t|\boldsymbol{R}-\boldsymbol{\lambda}'|^2/2} \le 1$: the same rule as before, with the trap centre shifted by f/k_t. Because $a \le 1$ for any f, the accepted chains are exact equilibrium samples *whatever* f *is*. The choice of f only affects efficiency.

Step 5: choose f. Acceptance is highest when proposed end points cluster around $\boldsymbol{\lambda}'$. We choose f such that the mean proposed end position equals it:

$$N\,\langle u_x\rangle_f = \lambda_x - \frac{f}{k_t}, \tag{38}$$

a one-dimensional root-finding problem (the left side increases with f, the right side decreases, so the root is unique). Physically f is the tension of the chain at that extension: the trap spring is stretched by f/k_t and pulls with force f, balancing the chain's entropic pull. With this choice the acceptance rate is about 1% at $N = 40$, 70% extension (instead of 10^{-9}), and 2–3% for $N = 10$. The remaining rejections come from the transverse and longitudinal spread of the proposed end point compared to the trap width $\sqrt{k_{\mathrm{B}}T/k_t}$.

Algorithm (`sampling.sample_positions`)

1. Solve $N\,\langle u_x\rangle_f = \lambda_x - f/k_t$ for f.
2. Draw a batch of chains: N bond lengths from $r\,g(r)I_0(\beta f r)$, angles from von Mises$(0, \beta f r)$.
3. Compute $\boldsymbol{R}$; keep each chain with probability $e^{-\beta k_t|\boldsymbol{R}-\boldsymbol{\lambda}'|^2/2}$.
4. Positions $\boldsymbol{r}_i = \sum_{j\le i}\boldsymbol{u}_j$; momenta Maxwellian; repeat until M chains are kept.

Unit tests check that tilted and untilted samplers give the same ensemble, and that the Gaussian chain reproduces its exact mean and covariance even at 70% extension.

10.3 Exact work statistics and dissipation estimate

Exact work statistics for $b = 0$. With linear forces, $(\boldsymbol{q}, \boldsymbol{p}, W)$ along x is a linear-Gaussian Markov chain (each sub-step is an affine map plus Gaussian noise). Propagating the mean and covariance gives the exact $\langle W\rangle$ and $\mathrm{Var}\, W$ of the *discrete* algorithm. The Jarzynski residual $\langle W\rangle - \beta\sigma_W^2/2 - \Delta F$ is $-8\times10^{-4}k_{\mathrm{B}}T$ at $\Delta t = 0.01$ and scales as Δt^2.

Dissipation estimate. In a slow pull the chain stretches affinely (bead i moves at $v\,i/N$), so the friction power is $\zeta N v^2/3$, and with $x_1 - x_0 = N/2$ and (9)

$$\beta W_{\mathrm{diss}} \simeq \frac{\zeta N (x_1-x_0)^2}{3k_{\mathrm{B}}T\tau} = \frac{\pi^2}{6}\frac{N}{\tau/\tau_1} \tag{39}$$

(the first version had $\pi^2/24$ because of the τ_1 error).

11 Design decisions (as implemented)

- Bath: underdamped Langevin with BAOAB.

- Both chains: $b = 0$ for exact validation, $b = 1$ for the physics.
- Sweeps at fixed reduced speed τ/τ_1, plus one fixed-τ temperature sweep.
- $k_\mathrm{B}T \in \{0.25, 0.5, 1, 2\}$, $N \in \{5, 10, 20, 40\}$, extension $0.2 \to 0.7$ of the contour length, $M = 10^5$ pulls per point (raised from the planned 10^4 because the GPU made it cheap).
- Crooks theorem and BAR: added after the first report (reverse pulls with the same τ).
- Initial states: exact (tilted) rejection sampling instead of equilibration runs.
- Corrections to the first version: τ_1 is $4\times$ shorter (clamped far end), hence the $\pi^2/6$ prefactor in (39). The rare-event criterion is $M \gtrsim e^{\beta W^R_\mathrm{diss}}$ (reverse dissipation), which reduces to $e^{\beta W_\mathrm{diss}}$ only for Gaussian work.

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